

$X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\underset{\uparrow}{\theta}, \sigma^2)$ θ : location param.

$$\hat{\theta} = \frac{1}{n} \sum x_i$$

$$\mathbb{E}(\hat{\theta}) = \theta$$

$$\mathbb{E}((\hat{\theta} - \theta)^2) = \text{Var}(\hat{\theta})$$

$$\begin{aligned} \mathbb{E}(\hat{\theta}) &= \frac{1}{n^2} \left[\sum \sigma^2 \right] = \frac{n\sigma^2}{n^2} \\ &= \frac{\sigma^2}{n} \end{aligned}$$

How optimal is this?

$$\inf_{\hat{\theta}} \sup_{\theta \in \mathbb{R}} \mathbb{E}((\hat{\theta} - \theta)^2) \geq \underbrace{\dots}_{O\left(\frac{\sigma^2}{n}\right)}$$

$$\Theta = \mathbb{R} \quad C = \{\theta^1, \dots, \theta^k\} \quad 2\delta\text{-Sep. Set} \quad Z, J \sim J \sim \text{Unif}(1 \dots k)$$

$$m \geq \frac{1}{\delta^2} \inf_{\psi} Q(\psi(Z) \neq J) \quad Z \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta^J, \sigma^2)$$

$$C = \{0, 2\delta\}$$

$$C = \{\theta_1, \theta_2\}$$

$$\downarrow Q(\psi(Z) \neq J) = P(J=1) \cdot Q(\psi(Z) \neq 1 | J=1) + P(J=2) \cdot Q(\psi(Z) \neq 2 | J=2)$$

$$\text{Let } A = \{Z \in \mathbb{R} \mid \psi(Z) = 1\}$$

$$= \frac{1}{2} \cdot P_{\theta_1}(\overbrace{\psi(Z) \neq 1}^{\bar{A}}) + \frac{1}{2} P_{\theta_2}(\overbrace{\psi(Z) \neq 2}^A)$$

$$= \frac{1}{2} P_{\theta_1}(\bar{A}) + \frac{1}{2} P_{\theta_2}(A)$$

$$= \frac{1}{2} [1 - P_{\theta_1}(A) + P_{\theta_2}(A)]$$

$$\inf_{\gamma} Q(\gamma(z) \neq J) = \inf_{A \subseteq \mathbb{R}} \frac{1}{2} [1 - \underbrace{P_{\theta_1}(A) + P_{\theta_2}(A)}_{g(A)}]$$

$$= \frac{1}{2} - \frac{1}{2} \sup_{A \subseteq \mathbb{R}} \underbrace{(P_{\theta_2}(A) + P_{\theta_1}(A))}_{-g(A)}$$

$$= \sup_{A \subseteq \mathbb{R}} |P_{\theta_1}(A) - P_{\theta_2}(A)|$$

$$=: \|P_{\theta_1} - P_{\theta_2}\|_{TV}$$

Total Variation
Distance.

$$= \frac{1}{2} - \frac{1}{2} \|P_{\theta_1} - P_{\theta_2}\|_{TV}$$

$$m \geq \frac{\sigma^2}{2} \left[1 - \|P_0^n - P_{2\delta}^n\|_{TV} \right] \\ \geq \frac{1}{2.12} \frac{\sigma^2}{n}$$

①

$$\|p - q\|_{TV} = \frac{1}{2} \int_{-\infty}^{+\infty} |p(x) - q(x)| dx$$

$$= \sup_{A \subseteq \mathbb{R}} \left| \int_A p(x) dx - \int_A q(x) dx \right|$$

$$A = \{x \mid p(x) > q(x)\} \\ = \frac{1}{2} \left(\int_{p>q} (p-q) dx + \int_{q>p} (q-p) dx \right) \\ = \frac{1}{2} \int_{-\infty}^{+\infty} |p-q| dx \\ \Rightarrow \bar{A}^* \text{ is also opt.}$$

Suppose A^* opt.

$$\int_{A^*} p dx - \int_{A^*} q < 0$$

Then

$$\int_{\bar{A}^*} p - \int_{\bar{A}^*} q \\ = \cancel{\int_{A^*} p} - \cancel{\int_{A^*} q} + \int_{\bar{A}^*} q \\ = \int_{\bar{A}^*} q - \int_{\bar{A}^*} p > 0$$

② Suppose p, q densities $p > 0$
 $q > 0$

$$\|p - q\|_{TV}^2 \leq \frac{1}{4} \left[\int \frac{p^2}{q} dx - 1 \right]$$

Ex. 15-10b

$$\|p - q\|_{TV}^2 = \frac{1}{4} \left(\int |p - q| dx \right)^2$$

$$= \frac{1}{4} \left(\int \underbrace{\frac{|p - q|}{\sqrt{q}}}_a \underbrace{\sqrt{q}}_b dx \right)^2$$

$$= \frac{1}{4} \cdot \int \frac{(p - q)^2}{q} dx \cdot \underbrace{\int q dx}_1 = \frac{1}{4} \int \frac{p^2 + q^2 - 2pq}{q} dx$$

$$= \frac{1}{4} \left[\int \frac{p^2}{q} dx + 1 - 2 \right]$$

$$\left[\underbrace{\int a(x) \cdot b(x)}_{\text{inner prod.}} \right]^2 \leq \underbrace{\int a^2(x) dx}_{\text{2-norm}} \cdot \int b^2(x) dx$$

(3) Extend this to $C = \{ \theta_1 = 0, \theta_2 = 2\delta \}$.

$$p = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{x^2}{2\sigma^2}\right\} \rightarrow p^n = \frac{1}{(\sqrt{2\pi}\sigma)^n} \exp\left\{-\frac{\|x\|^2}{2\sigma^2}\right\}$$

$$q = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{(x-2\delta)^2}{2\sigma^2}\right\} \rightarrow q^n = \frac{1}{(\sqrt{2\pi}\sigma)^n} \exp\left\{-\frac{\|x-2\delta\|^2}{2\sigma^2}\right\}$$

$$\begin{aligned} \|p^n - q^n\|_{TV}^2 &\leq \frac{1}{4} \left\{ \int \frac{p^{n2}}{q^n} dx - 1 \right\} \\ &= \frac{1}{4} \left\{ \int \frac{1}{(\sqrt{2\pi}\sigma)^n} \frac{\exp\left\{-\frac{2\|x-2\delta\|^2}{2\sigma^2}\right\}}{\exp\left\{-\frac{\|x\|^2}{2\sigma^2}\right\}} dx - 1 \right\} \\ &= \frac{1}{4} \left\{ \int \frac{\exp\left\{-\frac{(\|x\|^2 + 2\|\tilde{2\delta}\|^2 - 4x^T(\tilde{2\delta}))}{2\sigma^2}\right\}}{(\sqrt{2\pi}\sigma)^n} dx - 1 \right\} \\ &= \frac{1}{4} \left\{ \int \frac{\exp\left\{-\frac{\|x-2\delta\|^2}{2\sigma^2} + \frac{2\|2\delta\|^2 - 4x^T(2\delta)}{2\sigma^2}\right\}}{(\sqrt{2\pi}\sigma)^n} dx - 1 \right\} \end{aligned}$$

$$= \frac{1}{4} \left\{ e^{4\delta_n^2/\sigma^2} - 1 \right\}$$

$$m \geq \frac{\delta^2}{2} \left[1 - \frac{1}{2} \sqrt{e^{4n\delta^2/\sigma^2} - 1} \right] \quad \delta = \frac{1}{2} \frac{\sigma}{\sqrt{n}}$$

$$= \frac{\sigma^2}{8n} \left[1 - \frac{1}{2} \sqrt{e - 1} \right] \geq \frac{\sigma^2}{n} \times \frac{1}{24}$$

$0.344 > 1/3$

$$m \geq \frac{1}{24} \frac{\sigma^2}{n}$$

Le Cam's Method